

Garrett Fincke

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Education

University of Pittsburgh

Master of Science in Computer Science; SCI Scholarship for Excellence

Aug 2026 – Expected May 2028

Pittsburgh, PA

The Pennsylvania State University

Bachelor of Science in Computer Science

Aug 2021 – Dec 2024

University Park, PA

Experience

Westinghouse Electric Company

Software Developer Intern, GIC Tools and Applications

May 2026 – Dec 2026

Cranberry Township, PA

- Built and deployed a 15-operation Python/FastAPI service for Oracle-backed document data with filtering, reference resolution, and user-scoped access controls; validated deterministic retrieval across 500+ releases, 1,700+ hierarchy locations, and 60,000 configuration-item memberships
- Eliminated manual reconciliation by pairing the service with a refactored Prime client to flag revision mismatches and missing documents; parallelized throttled calls across 5–8 workers for 4–5x faster comparisons
- Integrating the document API with Westinghouse's Databricks agent platform to support bertha generative AI workflows; connecting it to an internal project database to automate stale-revision checks

Scale AI

Software Engineer (contract)

May 2024 – Jul 2025

Remote

- Built Python and Docker evaluation harnesses for LLM-generated code, automating compilation, execution, linting, testing, pass@k scoring, and failure analysis
- Created schema-versioned Pydantic and JSON Schema pipelines with deterministic sampling, replacing ad hoc manual review with reproducible validation and reducing evaluation turnaround time
- Built internal dashboards for evaluation failure analysis and PostgreSQL ingest checks for schema normalization and deduplication, improving data quality across annotation and QA workflows

Projects

SwimMate v2 | Python, Django REST, PostgreSQL/pgvector, OpenAI, Swift, SwiftUI, iOS/watchOS

Full-stack swimming platform with native iOS/watchOS apps and AI-generated workouts

- Built a Django/PostgreSQL RAG pipeline for personalized swim workouts with hybrid retrieval and reranking; added a deterministic 48-case eval suite tracking constraint pass rates, latency, and token usage
- Built SwiftUI iOS/watchOS apps with HealthKit workouts and offline sync; added token refresh and WatchConnectivity

Coral | TypeScript, React Ink, Ollama, Node.js, SQLite, TypeScript Language Server

Local-first CLI/TUI coding agent powered by Ollama with semantic retrieval and code intelligence

- Built a multi-turn coding agent with parallel tools, research subagents, semantic code search, MCP, and file, shell, and Git operations
- Improved local-model reliability through tool-call recovery, schema validation, loop detection, persistent SQLite sessions, conversation compaction, and TypeScript language intelligence

MDX Preview for VS Code | TypeScript, React 19, MDX 3, VS Code Extension API

Published VS Code/Open VSX extension for live MDX previews with 10,000+ Open VSX downloads

- Architected a six-package monorepo and published its reusable compiler/runtime as the mdx-forge npm package
- Built trust-gated Safe and Trusted rendering with DOMPurify, strict CSP, React component support, and compatibility shims for five MDX frameworks

Technical Skills

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, C/C++

Backend & Data: Django, FastAPI, Node.js, PostgreSQL/pgvector, Oracle, SQLite, Pydantic, REST/OpenAPI

AI/LLM: RAG, embeddings/vector search, agentic AI, MCP, Ollama, OpenAI/Anthropic APIs

Frontend & Mobile: React, Next.js, SwiftUI, UIKit, Tailwind CSS

Cloud & DevOps: Docker, Docker Compose, AWS EC2/SSM/CloudWatch, GitHub Actions, CI/CD, Vercel